Exploring the Trade-offs Between Audio Fidelity and Computational Efficiency in a Custom Real-Time Browser-Based Synthesizer

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How do different oscillator anti-aliasing techniques and digital filter architectures influence the relationship between audio fidelity and computational efficiency in real-time browser-based synthesis?

I. Literature Review

Digital signal processing (DSP) refers to the manipulation of signals represented in discrete form, such as those stored or processed by computers. In computer music, DSP provides the theoretical and practical foundation for digitally modelling and recreating the behaviour of analog audio systems, including oscillators and filters. However, accurately translating analog oscillators and filters into the digital domain presents several technical challenges. Specifically, the choice of oscillator anti-aliasing techniques and digital filter architectures introduces different considerations that influence the accuracy, stability, and computational efficiency of the resulting audio. These considerations become particularly important in real-time browser-based synthesis, where audio fidelity must be balanced against computational constraints.

This section examines these challenges through a review of existing literature, focusing on the methods or approaches used or proposed to address them.

A. Digital Oscillators

Analog oscillators generate the fundamental voltage signals used in analog synthesisers, while digital oscillators aim to reproduce equivalent waveforms in the discrete-time domain. Common waveforms include the sine, sawtooth, square and triangle waves, each with distinct harmonic characteristics that contribute to its perceived timbre.

A naive digital oscillator generates waveforms directly from mathematical functions or phase-based calculations. A sawtooth, for example, can be produced with a phase accumulator, which advances its phase on every sample by an amount determined by the oscillator frequency and the sample rate. The phase is mapped to an amplitude range and, once it reaches its maximum, wraps back to the minimum, producing the characteristic sawtooth shape. This abrupt jump is known as a *discontinuity*, and it introduces infinitely many high-frequency harmonics. The Nyquist–Shannon sampling theorem states that a signal can only be represented accurately if it contains no components above the Nyquist frequency, which is half the sampling frequency. A sawtooth always contains harmonics above this limit, so they cannot be represented and instead fold back into the audible range as inharmonic distortion known as aliasing.

The problem worsens as the fundamental frequency rises. This applies to any waveform with discontinuities, such as the sawtooth and also the square wave, because a greater proportion of their harmonics then lies above the Nyquist frequency.

B. Anti-Aliasing Techniques

Anti-aliasing refers to techniques that reduce the effect of aliasing. One approach is the Band-Limited Step (BLEP) method. The idea is to replace the abrupt discontinuity in the naive waveform with a band-limited approximation, which reduces the high-frequency content introduced by the discontinuity and so improves the accuracy of the eventual reconstruction.

An ideal step is the integral of an impulse, and an ideal band-limited step is therefore the integral of a band-limited impulse, typically a windowed sinc function[1]. Rather than synthesising the entire waveform from band-limited components, the BLEP method computes only the *residual*—the difference between the band-limited step and the naive (non-band-limited) step—and applies it as a correction to the naive waveform in the immediate vicinity of each discontinuity, reducing the effect of aliasing.

A widely used variant is the polynomial BLEP (PolyBLEP), which approximates the residual with a low-order polynomial instead of a table of sinc-derived values[2]. For a normalised phase $t \in [0, 1)$ and a per-sample phase increment $\Delta t = f_0/f_s$, where f_0 is the oscillator frequency and f_s the sampling frequency, a two-sample PolyBLEP residual is given by

$$r(t) = \begin{cases} 2\dfrac{t}{\Delta t} - \left(\dfrac{t}{\Delta t}\right)^2 - 1, & 0 \le t < \Delta t, \\ \left(\dfrac{t-1}{\Delta t}\right)^2 + 2\dfrac{t-1}{\Delta t} + 1, & 1 - \Delta t < t < 1, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

The corrected sawtooth is then obtained by subtracting this residual from the ideal but naive sawtooth waveform $s(t)$

$$y(t) = s(t) - r(t) = (2t - 1) - r(t), \tag{2}$$

which also smoothes the wrap-around over the samples adjacent to the discontinuity. The naive sawtooth wave is chosen to

be $s(t) = 2t - 1$ for simplicity of residual calculation and thus implementation of PolyBLEP, because it reaches its minimal extreme of $s(t) = -1$ at $t = 0$, and the maximum extreme of $s(t) = 1$ at $t = 1$; the discontinuities exist at integer intervals.

A square wave may be treated as the sum of two sawtooth-like discontinuities, one rising and one falling, so the same residual is applied at both transitions (with opposite sign). A triangle wave, being continuous, has no step discontinuities, but its first derivative is discontinuous; the slope discontinuity can be corrected with an analogous polynomial band-limited ramp (PolyBLAMP) residual[3]

The principal advantages of PolyBLEP and PolyBLAMP are their low computational cost, the lack of lookup tables, and suitability for real-time use with modulated frequency, since the correction depends only on the current phase and phase increment. Its limitation is that a low-order polynomial only approximates the ideal band-limited step, so some residual aliasing remains, particularly at high fundamental frequencies, where it is attenuated rather than eliminated.

Alternative approaches offer different trade-offs. Oversampling reduces aliasing by generating waveforms at higher sampling rates before filtering and decimation, but increases computational cost [4]. Wavetable synthesis uses precomputed band-limited waveforms to suppress harmonics above the Nyquist frequency, but requires additional memory and table selection, reducing flexibility especially during dynamic modulation[5].

C. Digital Filter Design Architectures

Filters shape the harmonic content of an oscillator's output. A lowpass filter attenuates components above a cutoff frequency, a highpass filter attenuates those below it, and a bandpass filter passes a band around the cutoff while attenuating components on either side. In a synthesizer, the cutoff frequency and resonance are typically modulated in real time, so a digital filter must remain stable and accurate while its coefficients change from sample to sample.

The biquad is the most widely used digital filter structure. It is a second-order infinite impulse response (IIR) filter with a transfer function

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}, \tag{3}$$

whose coefficients are computed from the cutoff frequency, resonance and filter type, commonly using the closed-form expressions of the Audio EQ Cookbook[**?**]. Its advantages include low computational cost and the flexibility to obtain different filter responses through changes to the numerator coefficients.

However, its conventional difference-equation implementation presents some challenges under rapid real-time parameter modulation. Changes to the cutoff frequency or resonance require recalculation of the filter coefficients, while the internal delay states retain values generated under the previous coefficient configuration. This mismatch can introduce transient artefacts during modulation[**?**].

To address some of these limitations, the State Variable Filter (SVF) can be implemented using a zero-delay feedback (ZDF) topology and a topology-preserving transform (TPT). Rather than implementing the filter as a conventional difference equation, this structure uses integrator-equivalent states to calculate the lowpass, bandpass and highpass responses. Specifically, the trapezoidal integration method—a known mathematical technique for approximating integrals—allows the feedback structure to be discretised without introducing an additional unit-sample delay into the feedback path. This provides favourable numerical behaviour and makes the structure particularly suitable for real-time modulation of cutoff frequency and resonance. To be clear, the ZDF SVF remains an IIR filter, but its topology offers advantages over conventional biquad implementations when filter parameters vary rapidly.

The Moog ladder filter is a fourth-order lowpass structure comprising four cascaded one-pole stages with a global feedback path from the output to the input, which produces the resonance and the characteristic $24\,\mathrm{dB}$ per octave roll-off. Because the feedback loop contains four stages, a naive digital implementation that inserts a unit delay into the loop, as in early models such as that of Stilson and Smith, loses accuracy at high cutoff frequencies. The resonance is also coupled to the cutoff frequency, and the filter can become unstable at high resonance settings. In addition, only the lowpass response is available directly, so highpass and bandpass responses must be derived by combining the outputs of individual stages. The ZDF ladder resolves the delay-free loop by solving the nonlinear or linearised feedback equation for the current sample, again using trapezoidal integration, which preserves the cutoff and resonance behaviour of the analog circuit and improves stability, at a higher computational cost than the IIR ladder owing to the additional algebra, and potentially iterative solving when saturation is modelled[**?**].

Taken together, these topologies represent a trade-off between computational cost, stability and fidelity to the analogue response. The biquad is the cheapest but the least robust under modulation. The ZDF SVF offers a favourable balance, with accurate responses across all three filter modes at moderate cost. The Moog ladder offers a distinctive resonant character but at the greatest cost and complexity, and its ZDF form improves accuracy and stability over the IIR form at a further increase in computation.

II. Methods

hello world

III. Results

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IV. Discussion

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V. Conclusion

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